

LARS JANSEN

Senior Software Engineer

Assen, Drenthe, Netherlands | +31 97058049397 | larsjansen0521@outlook.com | linkedin.com/in/lars-jansen-0a37b8393

PROFESSIONAL SUMMARY

Senior Mobile Engineer with **9+ years of software engineering experience** building large-scale mobile and cloud-connected platforms across logistics, SaaS, and enterprise analytics industries. Specialized in **Flutter (2.x–3.x)** and **Native Android with Kotlin (1.5–1.9) and Java (8–17)** to deliver **high-performance cross-platform applications** used by thousands of active users. Strong background in **object-oriented architecture, scalable UI systems, and performance optimization**, including solving memory leaks, asynchronous rendering issues, and API latency problems in complex mobile environments. Experienced migrating legacy **native Android codebases into Flutter modular architecture**, integrating **AI-assisted development tools**, and delivering adaptive mobile interfaces with robust offline capability, real-time data synchronization, and resilient network handling for mission-critical applications.

SKILLS

- **Mobile Development:** Flutter (2.x–3.x), Dart (2.x–3.x), Android SDK, Kotlin (1.5–1.9), Java (8–17), Material UI systems, adaptive layouts, mobile state management
- **Architecture & Programming:** Object-Oriented Programming, Clean Architecture, MVVM, Repository pattern, dependency injection, modular mobile architecture
- **Backend Integration:** REST APIs, JSON serialization, asynchronous networking, caching layers, mobile data synchronization
- **Performance & Optimization:** rendering optimization, memory profiling, frame drop reduction, async task optimization, network performance tuning
- **Tools & Collaboration:** Git, CI/CD pipelines, AI-assisted development tools, mobile debugging tools, Agile collaboration
- **Platforms & Integrations:** Android ecosystem, wearable integrations, device communication APIs, IoT device connectivity patterns

WORK HISTORY

Senior Software Engineer RavenTrack

August 2021 to November 2025

- Led the architecture and development of a **large-scale logistics tracking mobile platform using Flutter (3.x) and Dart**, delivering cross-platform Android applications that processed thousands of real-time location updates while maintaining smooth UI rendering and efficient background synchronization.
- Resolved a complex **Flutter rendering performance issue** where map widgets caused frame drops during high-frequency GPS updates by optimizing widget rebuild cycles and implementing selective state updates, reducing UI latency and stabilizing runtime performance.
- Migrated a legacy **Native Android application written in Java (8–11)** to a modular **Flutter (2.x → 3.x) architecture**, gradually replacing old activities with reusable Flutter components while maintaining compatibility with existing backend APIs.
- Implemented a scalable mobile networking layer integrating REST APIs with asynchronous Dart isolates, which solved intermittent request blocking issues that previously caused UI freezes during high-traffic mobile usage.
- Designed adaptive UI components using **Flutter Material widgets**, enabling consistent responsive layouts across a wide range of Android device screen sizes while maintaining accessibility and usability standards.
- Identified and fixed a persistent **memory leak caused by improper lifecycle management in asynchronous network listeners**, resolving crashes reported by QA during extended background usage.
- Integrated **AI-assisted development tooling** to automate repetitive UI refactoring tasks and code quality analysis, improving development velocity while maintaining strict performance and reliability requirements.
- Built advanced offline-first data synchronization logic that allowed field users to continue operating the mobile system in low-connectivity environments while safely reconciling updates when connectivity was restored.
- Implemented mobile analytics instrumentation to trace slow rendering paths and backend response latency, which helped identify inefficient API endpoints causing delayed data rendering.
- Collaborated closely with backend engineers to redesign inefficient mobile endpoints that originally returned excessive payloads, significantly improving mobile response time and reducing bandwidth usage.
- Introduced structured error handling and centralized logging for mobile exceptions, making production debugging significantly easier and improving long-term maintainability of the application.
- Optimized background location processing and battery consumption by restructuring periodic update logic and improving task scheduling behavior on Android devices.
- Mentored engineers in **Flutter architecture, Dart best practices, and mobile performance tuning**, helping the team transition from native Android development to a modern cross-platform mobile stack.

Software Engineer Infinite Loop

June 2018 to July 2021

- Developed mobile features using **Flutter (2.x)** and **Native Android Kotlin (1.5+)**, building scalable UI modules that interacted with backend SaaS services while maintaining strict performance and reliability standards.
- Resolved a recurring **application crash triggered by concurrent API calls in Kotlin coroutines**, introducing structured concurrency patterns that stabilized network execution and eliminated race condition errors.

- Implemented reusable UI components in Flutter that improved code reuse across several mobile screens and reduced feature development time while maintaining consistent user experience design.
- Migrated older Android UI screens implemented with Java activities into Flutter widgets while preserving backend integration and ensuring smooth navigation between native and Flutter modules.
- Designed a structured mobile API client that handled retries, caching, and network failures gracefully, preventing data corruption and improving reliability for high-traffic mobile sessions.
- Introduced automated mobile build pipelines that improved release reliability and reduced manual deployment errors in the engineering workflow.
- Collaborated with product teams to design flexible UI flows that adapted to changing business requirements without requiring major architecture rewrites.
- Improved mobile loading performance by restructuring inefficient API response parsing and reducing heavy object allocations during application startup.
- Integrated device communication features enabling mobile apps to interact with connected hardware devices using Android system APIs.
- Performed extensive debugging and profiling to identify performance bottlenecks affecting scrolling and screen transitions in complex UI views.
- Implemented improved state management patterns that significantly reduced unnecessary UI rebuilds and improved overall application responsiveness.

Software Developer Datamart

October 2015 to May 2018

- Built early mobile analytics and internal tools using **Native Android Java (8)**, designing structured application modules that interacted with enterprise backend reporting systems.
- Resolved synchronization issues between mobile data caches and backend SQL reporting services by implementing improved request scheduling and background update mechanisms.
- Designed modular Android UI screens that enabled data visualization for operational analytics dashboards used by internal teams.
- Implemented asynchronous networking patterns that prevented UI blocking when retrieving large data sets from backend reporting APIs.
- Optimized slow mobile queries by restructuring data payload formats and implementing efficient client-side parsing strategies.
- Debugged memory pressure issues caused by large in-memory datasets and implemented pagination logic to prevent application crashes on lower-memory devices.
- Collaborated with backend engineers to redesign inefficient endpoints that previously caused excessive network latency in mobile dashboards.
- Improved reliability of background update services that synchronized operational metrics from remote systems into the mobile application.
- Implemented structured logging and diagnostic tools to help identify production issues reported by enterprise users.

EDUCATION

Bachelor's degree in Computer Science University of Amsterdam

2011 to 2015

- Built strong foundations in software engineering, data structures, and systems design, with coursework that supports designing reliable services and data-driven applications.

PROJECTS

Real-Time Logistics Tracking Platform

- Developed a high-performance mobile logistics tracking platform using **Flutter and Dart**, enabling field teams to monitor real-time shipment locations while maintaining smooth UI performance under heavy location data updates.
- The project included solving complex synchronization challenges between mobile devices and backend services while ensuring reliable offline functionality and efficient background data processing.